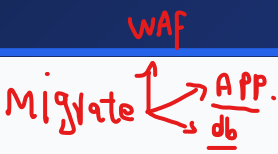


Stage 1: On-Premises Infrastructure Setup & Landing Zone Deployment Guide

Complete Step-by-Step Manual Guide: Hyper-V Nested Virtualization, 4-Tier Application Hosting, and Target Azure Landing Zone Provisioning



1. Executive Summary & Lab Aims

The objective of **Stage 1** is to establish a fully functional, multi-tier legacy on-premises environment (the **SmartHotel360 Application**) hosted inside an Azure Virtual Machine via **Nested Virtualization (Hyper-V)**, alongside a pre-configured **Azure Target Landing Zone**.

Before conducting discovery, assessment, and migration using **Azure Migrate** in Stage 2, this stage builds the source infrastructure from scratch, including custom virtual networks, internal NAT gateways, load balancing/WAF proxies, web application servers, and backend database engines.

Key Architectural Goal

Simulate an authentic corporate datacenter hosting the SmartHotel web application across 4 distinct VMs, while exposing public access via Port Forwarding/NAT and provisioning the destination Azure PaaS/IaaS resources in resource group SmartHotel-RGRG.

2. On-Premises Architecture & VM Specifications

The simulated on-premises infrastructure consists of one physical/parent hypervisor VM (SmartHotelHost) running Windows Server and Hyper-V. Inside this host, 4 nested guest virtual machines are connected via an Internal Virtual Switch (SmartHotelSwitch) with a NAT subnet of 192.168.10.0/24.

VIRTUAL MACHINE	ROLE / TIER	OPERATING SYSTEM	IP ADDRESS	KEY SERVICES / SOFTWARE INSTALLED
SmartHotelHost	Parent Hyper-V Host	Windows Server 2019/2022	10.0.0.4 (Azure VNet)	Hyper-V Role, NAT Gateway (192.168.10.1)
UbuntuWAF	Web Proxy / WAF Tier	Ubuntu 18.04 / 20.04 LTS	192.168.10.10	NGINX Reverse Proxy & Load Balancer (Port 80)
SmartHotelWeb1	Web Tier (Frontend)	Windows Server 2012 R2	192.168.10.11	IIS Web Server, ASP.NET Frontend Application UI
SmartHotelWeb2	Application Tier (Backend)	Windows Server 2012 R2	192.168.10.12	IIS Web Server, ASP.NET Web API / Services
SmartHotelSQL1	Database Tier	Windows Server 2016	192.168.10.20	SQL Server 2017 (Developer/Express), TCP 1433

3. Detailed Implementation Guide: On-Premises Hyper-V Host

1 Provision SmartHotelHost VM in Azure

Create the primary Hyper-V host machine in Azure under Resource Group SmartHotel-RGHostRG:

- **VM Name:** SmartHotelHost | **Region:** East US
- **OS Image:** Windows Server 2019 or 2022 Datacenter
- **Size:** Standard_D4s_v3 (4 vCPUs, 16 GB RAM) — *Must support Nested Virtualization.*
- **Security Type:** Standard (Do NOT use Trusted Launch, as it blocks nested virtualization extensions).
- **Disks:** Attach an OS Disk (SmartHotelHost_OsDisk) and a secondary 256 GB Data Disk (SmartHotelHost_DataDisk) formatted as drive E: for nested VM storage.

2 Configure Hyper-V, Virtual Switch, and NAT Networking

RDP into SmartHotelHost using its Public IP (smarthotelhostip). Open PowerShell as Administrator and run:

```
# Step A: Install Hyper-V Feature
Install-WindowsFeature -Name Hyper-V -IncludeManagementTools -Restart

# (After host reboots, log back in via RDP)

# Step B: Create Internal Hyper-V Switch
New-VMSwitch -Name "SmartHotelSwitch" -SwitchType Internal

# Step C: Assign Gateway IP to host virtual interface
New-NetIPAddress -IPAddress 192.168.10.1 -PrefixLength 24 -InterfaceAlias "vEthernet
(SmartHotelSwitch)"

# Step D: Configure NAT Gateway for nested VMs internet outbound connectivity
New-NetNat -Name "SmartHotelNAT" -InternalIPInterfaceAddressPrefix "192.168.10.0/24"
```

3 Configure Public Access via NAT Port Forwarding

Allow external internet users to access the SmartHotel web application dashboard through the Azure Public IP of SmartHotelHost by forwarding inbound Port 80 traffic to UbuntuWAF (192.168.10.10).

```
# Enable Static NAT Port Mapping on SmartHotelHost
Add-NetNatStaticMapping -NatName "SmartHotelNAT" -Protocol TCP -ExternalIPAddress 0.0.0.0 -
ExternalPort 80 -InternalIPAddress 192.168.10.10 -InternalPort 80

# Open Windows Firewall Port 80 on SmartHotelHost
New-NetFirewallRule -DisplayName "Allow HTTP Port 80 Inbound" -Direction Inbound -LocalPort 80 -
Protocol TCP -Action Allow
```

4. Detailed Configuration of Nested Guest Virtual Machines

A. Database Server Configuration (SmartHotelSQL1)

- **Network Settings:** IP: 192.168.10.20, Subnet Mask: 255.255.255.0, Gateway: 192.168.10.1, DNS: 8.8.8.8.
- **SQL Server Installation:** Install SQL Server 2017. Select Mixed Mode Authentication (SQL Server and Windows Authentication) and set the sa account password.

- **Database Provisioning:** Create database SmartHotel. Seed tables: Customers, Bookings, Rooms, and CheckIns.
- **Network & Firewall Protocols:**
 1. Open *SQL Server Configuration Manager* > Enable **TCP/IP** under *SQL Server Network Configuration*.
 2. In Windows Firewall, allow inbound TCP Port 1433:

```
New-NetFirewallRule -DisplayName "SQL Server TCP 1433" -Direction Inbound -Protocol TCP -
LocalPort 1433 -Action Allow
```

B. Web & Application Servers Configuration (SmartHotelWeb1 & SmartHotelWeb2)

- **SmartHotelWeb1 IP:** 192.168.10.11 | **SmartHotelWeb2 IP:** 192.168.10.12
- **Roles & Features:** Install **Web Server (IIS)** and **ASP.NET 4.7 / .NET Core Hosting Bundle**.
- **Application Deployment:** Extract SmartHotel web application source files to C:\inetpub\wwwroot\SmartHotel.
- **Database Connection Configuration:** Update web.config to connect directly to SmartHotelSQL1:

```
<connectionStrings>
  <add name="DefaultConnection"
        connectionString="Data Source=192.168.10.20;Initial Catalog=SmartHotel;User
ID=sa;Password=YourPassword123!;"
        providerName="System.Data.SqlClient" />
</connectionStrings>
```

C. Reverse Proxy & WAF Configuration (UbuntuWAF)

- **Network Settings:** IP: 192.168.10.10, Gateway: 192.168.10.1, DNS: 8.8.8.8.
- **NGINX Installation & Setup:**

```
sudo apt update && sudo apt install nginx -y
sudo rm /etc/nginx/sites-enabled/default
sudo nano /etc/nginx/sites-available/smarthotel.conf
```

- **Reverse Proxy Configuration File (smarthotel.conf):**

```
upstream smarthotel_backend {
    server 192.168.10.11:80; # SmartHotelWeb1
    server 192.168.10.12:80; # SmartHotelWeb2
}

server {
    listen 80;
    server_name _;

    location / {
        proxy_pass http://smarthotel_backend;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

- **Enable & Reload NGINX:**

```
sudo ln -s /etc/nginx/sites-available/smarthotel.conf /etc/nginx/sites-enabled/  
sudo nginx -t  
sudo systemctl reload nginx
```

5. Azure Target Landing Zone Breakdown (Resource Group: SmartHotel-RGRG)

The resource group SmartHotel-RGRG acts as the modern Azure Target Landing Zone. It holds the destination services where workloads and databases will be migrated and modernized during Stage 2 and Stage 3.

RESOURCE NAME	RESOURCE TYPE	ARCHITECTURAL PURPOSE IN LANDING ZONE
SmartHotelVNet	Virtual Network	Target Cloud Network (10.1.0.0/16) with subnets for App, WAF, and Bastion.
SmartHotel-WAF	Application Gateway (WAF v2)	Replaces UbuntuWAF VM with cloud-native Layer-7 load balancing and security protection.
SmartHotel-WAF-ip	Public IP Address	Public IP attached to Azure Application Gateway for web app access.
smarthotelldb	Azure SQL Database	Target PaaS database replacing SmartHotelSQL1 for database modernization.
smarthotelsqlggjxc	SQL Logical Server	Logical server hosting the smarthotelldb PaaS instance in Azure East US.
SmartHotelBastion	Azure Bastion	Provides secure RDP/SSH portal access to migrated VMs without public IPs.
SmartHotelBastion-ip	Public IP Address	Dedicated Public IP assigned to Azure Bastion service.
SmartHotelLA	Log Analytics Workspace	Centralized logging and diagnostic monitoring for Azure Migrate & migrated resources.

6. Stage 1 Operational Verification Checklist

Verify Complete Setup Before Moving to Stage 2 (Lab 2):

- Public Web Access:** Open a browser and navigate to `http://<smarthotelhostip>`. Confirm that the SmartHotel dashboard page loads showing guest check-in lists.
- Internal Connectivity:** Ping `192.168.10.1` from all 4 nested VMs to confirm gateway communication.
- Database Connection:** Confirm web pages fetch live customer records from `192.168.10.20` SQL Server.
- Azure Target Preparedness:** Confirm SmartHotel-RGRG contains active instances of SmartHotelVNet, SmartHotel-WAF, and smarthotelldb.